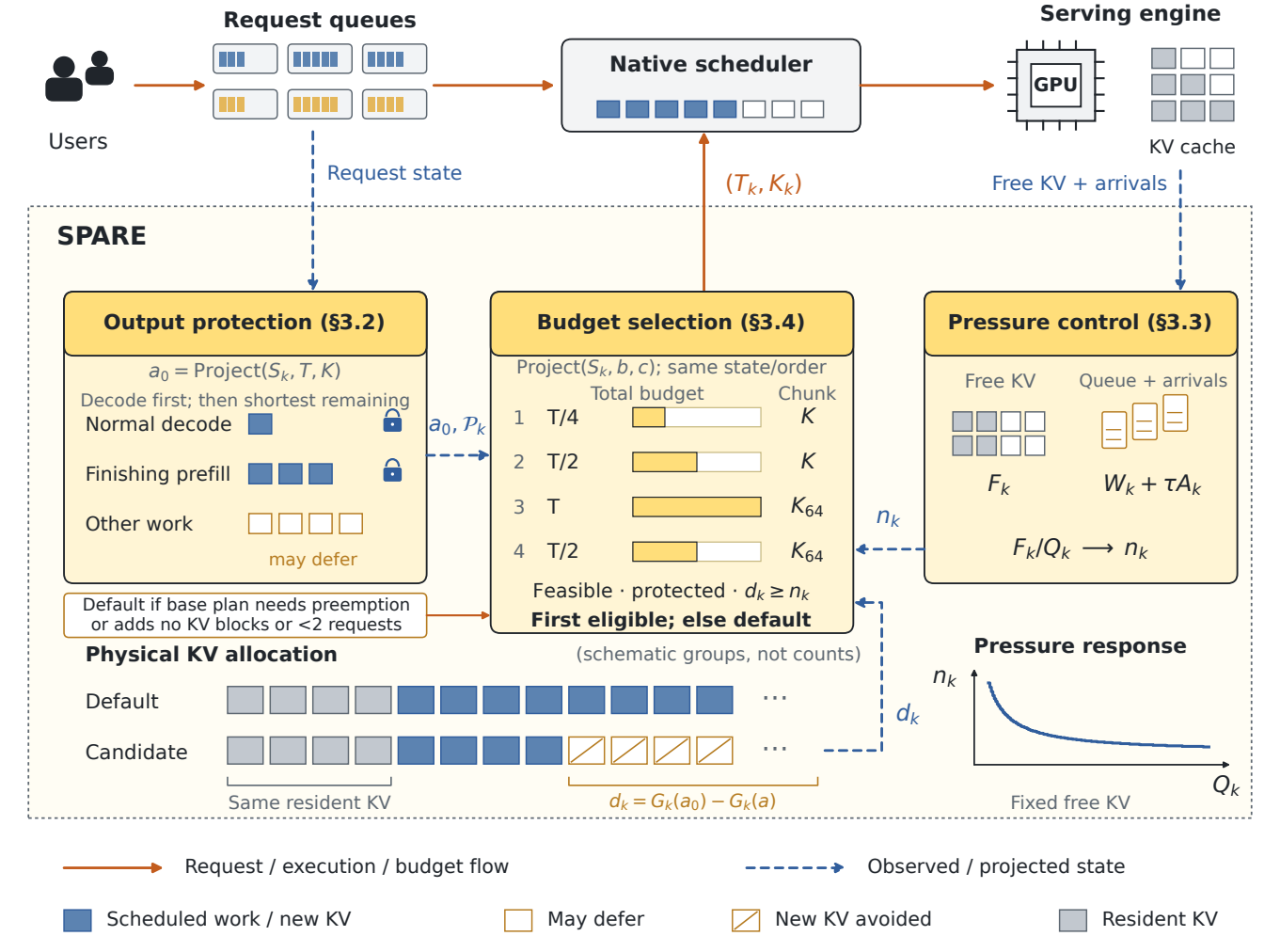
Request queues
Serving engine
Users
Native scheduler
GPU
KV cache
Request state
(T_k, K_k)
Free KV + arrivals
SPARE
Output protection (§3.2)
$a_0 = \text{Project}(S_k, T, K)$
Decode first; then shortest remaining
Normal decode
Finishing prefill
Other work
may defer
$a_0, \mathcal{P}_k$
Budget selection (§3.4)
Project(S_k, b, c); same state/order
Total budget
Chunk
1 T/4 K
2 T/2 K
3 T K_{64}
4 T/2 K_{64}
Feasible · protected · $d_k \geq n_k$
First eligible; else default
Pressure control (§3.3)
Free KV
Queue + arrivals
F_k
$W_k + \tau A_k$
$F_k/Q_k \longrightarrow n_k$
n_k
Default if base plan needs preemption or adds no KV blocks or <2 requests
Physical KV allocation
(schematic groups, not counts)
Default
Candidate
d_k
Same resident KV
$d_k = G_k(a_0) - G_k(a)$
Pressure response
n_k
Q_k
Fixed free KV
Request / execution / budget flow
Observed / projected state
Scheduled work / new KV
May defer
New KV avoided
Resident KV